



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	MATERIALS SCIENCE				
Type of Course	Major Elective (SPECIALIZATION I: MATERIALS SCIENCE)				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	A strong foundation in physics and chemistry.				
Course Summary	This course aims to provide students with a comprehensive understanding of the fundamental principles underlying the behavior of materials, as well as the cutting-edge technologies driving innovation in this field.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the various types of materials, the bonding between the elements and molecules, and type of interacting forces among the molecular systems	U, An	F	Instructor-created exams / Quiz
CO2	Develop a fundamental understanding of the importance of the structure of the compounds and performance of materials.	U,R	F	Instructor-created exams / Quiz

CO3	Gain knowledge about the different types of materials that are used in different applications and the different properties of diversified materials.	U, Ap	F	Instructor-created exams / Quiz
CO4	Familiarize students with advanced characterization techniques used to analyze materials structurally, surface, optically, electrically and magnetically.	U, An	F	Instructor-created exams / Quiz
CO5	Explore the applications of advanced materials in various industries, energy technology, and electronic and other applications..	U, An	F	Instructor-created exams / Quiz
CO6	Make the students capable of developing various materials through project work.	U, Ap	F	Instructor-created exams / Quiz
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	MATERIALS, INTERATOMIC FORCES, AND BONDING IN SOLIDS		10	16
	1	What is material science and need of material science (Elementary ideas only)	1	
	2	Classification of materials – metals, ceramics, polymers, composites, Advanced materials, need of modern material	2	
	3	Bonding forces and energies, Primary Interatomic Bonds	2	
	4	Ionic Bonding, Covalent Bonding, Metallic Bonding, van der Waals Bonding	3	
	5	Examples of anomalous volume expansion of water	2	
	Sections 1.2 – 1.6, 2.5 – 2.8 of Book 1			

II	Crystal Structure and Imperfections in Solids		10	16
	6	Single Crystals, Polycrystalline materials, Anisotropy, Nanocrystalline solids	3	
	7	Imperfections, Vacancies and Self Interstitials	2	
	8	Impurities in Solids, Specification of composition, Dislocations-Linear defects, Interfacial defects, Volume defects	3	
	9	Atomic Vibrations, Microstructure, Grain size determination.	2	
Sections 3.13 to 3.17, 4.1 to 4.11 of Book 1				
III	TYPES OF MATERIALS		17	22
	10	Conductors, insulators, and dielectrics: Thermal conductivity and electrical resistivity	3	
	11	Drilling down: the origins and manipulation of electrical properties,	3	
	12	Magnetic Materials: the physics and manipulation of magnetic properties	2	
	13	Materials selection for magnetic design	2	
	14	Materials for optical devices: The interaction of materials and radiation, the physics and manipulation of optical properties	4	
	15	The durability of Materials: oxidation, corrosion, and degradation	3	
Sections 14.1 – 14.4, 15.1 – 15.4, 16.1 – 16.4, 17.1 – 17.2 of Book 2				
IV	CHARACTERIZATION STUDIES AND TECHNIQUES		11	16
	16	Electrical and electronic measurements	2	
	17	Hall Effect in Semiconductors Introduction,	2	
	18	Magnetism and Magnetic Measurement	1	
	19	Introduction, Electrochemical Techniques	2	
	20	Introduction, Cyclic Voltammetry	1	
	21	Optical Microscopy, Photoluminescence Spectroscopy	2	
	22	Raman Spectroscopy of Solids.	1	
	Relevant sections from Book 3			
V	OPEN ENDED MODULE		12	

		Synthesis of Gold / Silver Nanoparticle and Introductory Soft Lithography Using PDMS		
		Thin film deposition by spin coating or Dip coating or spray pyrolysis techniques (Metal Oxides: any compound)		
		Solid State Reaction of Powder Ceramics		
		Any two Sections from the Chapter 8/9/10 of book 4, Book 5 or reference 6		

Books and References:

1. Materials Science and Engineering An Introduction, 7th Edition by William D. Callister, Jr, John Wiley & Sons, Inc (Book 1)
2. Materials Engineering, Science, Processing and Design: Michael Ashby, Hugh Shercliff and David Cebon, Published by Elsevier Ltd (Book 2)
3. Characterization of Materials: Elton N. Kaufmann, Volumes 1 and 2, John Wiley and Sons Publications, 2023 (Book 3)
4. Nanotechnology: Principles and Practices, Sulabha K. Kulkarni, Springer, 3rd Edition (Book 4)
5. Simple Chemical Methods for Thin Film Deposition: Synthesis and Applications, Springer, ISBN 978-981-99-0960-5 (Book 5)
6. Journal of Materials Science and Technology, 2013, 29 (5), 419 - 422

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	2	2	1	3	3	2	2	3	2	2	3	2
CO 2	3	1	3	2	3	3	3	2	3	2	3	3	3
CO 3	3	2	3	1	3	3	3	2	3	2	2	3	2
CO 4	3	1	2	2	3	3	2	2	3	2	3	2	0
CO 5	3	2	3	2	3	3	3	2	3	2	3	3	2
CO 6	3	0	1	1	3	3	1	2	3	2	3	3	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	